

Finite-size scaling with GP scaling functions and FSS mixture discrepancy (κ, ω)

1 Overview

Extracting critical exponents from simulations of statistical field theories is traditionally done with a complicated, but ad hoc, frequentist procedure. The goal of this project, in the spirit of [1], is to systematize that procedure using Bayesian probability.

2 Finite-size scaling from the renormalization group

View renormalization as an operator $R_b : K \rightarrow K$ (for $b \in \mathbb{R}$) on the coupling space of a theory K . We want the eigenvalues of the linearization J_b of R_b around a fixed point of interest. If

$$J_b(u_i) = b^{y_i} u_i, \quad (1)$$

then y_i is the RG eigenvalue of the corresponding scaling field u_i (often written b^{-x_i} for the dual operator scaling dimensions; we keep the y_i convention below).

For an observable O , define the pushforward $R_b^* O = O \circ R_b$. This likewise has an eigendecomposition. For an eigenoperator Φ of scaling dimension x_O ,

$$\langle \Phi \rangle(t, \{u_i\}) = b^{-x_O} \langle \Phi \rangle(b^{y_t} t, \{b^{y_i} u_i\}). \quad (2)$$

Taking the finite-size identification $b = L$,

$$\langle \Phi \rangle(t, \{u_i\}) = L^{-x_O} \langle \Phi \rangle(t L^{y_t}, \{u_i L^{y_i}\}). \quad (3)$$

A general observable is a linear combination of such eigenoperators, so

$$\langle O \rangle_L = L^{-x_0} \left[F_0(\{u_i L^{y_i}\}) + \sum_{a>0} L^{-(x_a - x_0)} F_a(\{u_i L^{y_i}\}) \right] \quad (4)$$

where x_0 is the leading (most relevant) operator dimension contributing to O , and the sum collects subleading operators.

2.1 Ising magnetization moments

Specialize to the 2D Ising universality class with a single relevant thermal perturbation. Write

$$t = \frac{T - T_c}{T_c} \quad (5)$$

for the reduced temperature, so the thermal scaling field is $u_t \propto t$ with RG eigenvalue

$$y_t = \frac{1}{\nu}. \quad (6)$$

The leading irrelevant bulk operator has eigenvalue $y_{\text{irr}} = -\omega$ (Wegner's correction-to-scaling exponent), and we write u for its (nonlinear) scaling field. Keeping only these two fields, the arguments of the scaling functions are

$$z := t L^{1/\nu}, \quad u L^{-\omega}. \quad (7)$$

For the absolute magnetization density $m = \langle |M| \rangle / N$, the leading magnetic scaling dimension is $x_0 = \beta/\nu$, so

$$m(t, L) = L^{-\beta/\nu} \left[F_m(z, u L^{-\omega}) + O(L^{-(x_1 - \beta/\nu)}) \right]. \quad (8)$$

The moments $\langle m^2 \rangle$ and $\langle m^4 \rangle$ are controlled by 2 and 4 powers of the magnetic operator, hence

$$m^2(t, L) = L^{-2\beta/\nu} \left[F_{m^2}(z, u L^{-\omega}) + \dots \right], \quad (9)$$

$$m^4(t, L) = L^{-4\beta/\nu} \left[F_{m^4}(z, u L^{-\omega}) + \dots \right]. \quad (10)$$

(Equivalently: collapsing with $L^{k\beta/\nu}$ for the k -th moment removes the leading L -power and leaves a universal function of $(z, u L^{-\omega})$.)

Expanding the leading scaling function in the irrelevant direction near the fixed point,

$$F_c(z, u L^{-\omega}) = f_c(z) + (u L^{-\omega}) g_c(z) + O(L^{-2\omega}), \quad (11)$$

so the collapsed fields

$$\Phi_m := m L^{\beta/\nu}, \quad \Phi_{m^2} := m^2 L^{2\beta/\nu}, \quad \Phi_{m^4} := m^4 L^{4\beta/\nu} \quad (12)$$

satisfy, to leading correction,

$$\Phi_c(z, L) = f_c(z) + u L^{-\omega} g_c(z) + \dots. \quad (13)$$

Thus the natural finite-size correction to a collapsed observable is multiplicative in $L^{-\omega}$: that is why the Wegner factor appears as a power of L , not of t alone.

2.2 Why $a = L^{-\omega} + \kappa |t|^{\omega\nu}$

Equation (13) already motivates an amplitude proportional to $L^{-\omega}$ at $t = 0$. Away from criticality the irrelevant scaling field itself depends on t , and bulk Wegner corrections produce terms $\sim |t|^\Delta$ with $\Delta = \omega\nu$. A minimal two-term ansatz that (i) matches the finite-size correction $L^{-\omega}$ and (ii) matches the bulk correction $|t|^{\omega\nu}$, while remaining homogeneous under the collapse $(t, L) \mapsto z$, is

$$a(t, L) = L^{-\omega} + \kappa |t|^{\omega\nu} = L^{-\omega} (1 + \kappa |z|^{\omega\nu}). \quad (14)$$

The factored form makes the RG content explicit: the whole correction is $O(L^{-\omega})$ times a smooth function of the collapse coordinate z , and $1 + \kappa |z|^{\omega\nu}$ is a simple positive surrogate for that z -dependence (with relative weight κ). In particular:

- $L^{-\omega}$ comes from the RG eigenvalue $y_{\text{irr}} = -\omega$ of the leading irrelevant operator in (4) and (11);
- $|t|^{\omega\nu}$ is the bulk conjugate ($|t|^\Delta$ with $\Delta = \omega\nu$), equivalently $|z|^{\omega\nu} L^{-\omega}$ after collapse;
- $a \rightarrow 0$ whenever $t \rightarrow 0$ and $L \rightarrow \infty$ together, so the asymptotic scaling window is recovered.

3 The Bayesian setup

As in any Bayesian problem, we start from unknowns and knowns. Concretely, for the 2D Ising model with one relevant thermal perturbation:

Unknowns. The critical parameters $\theta = (T_c, \nu, \beta)$ (and the correction exponents / amplitudes ω, κ), together with the universal scaling functions f_c (and discrepancy functions g_c) for each observable channel $c \in \{m, m^2, m^4\}$.

Knowns.

- Critical exponents are positive; T_c is positive.
- Scaling functions are smooth, with a characteristic length scale in z that we encode through a GP kernel.
- We can estimate observables at any (T, L) by Monte Carlo, up to a statistical uncertainty σ ; reducing σ is costly and compute is finite.
- For fixed θ , the collapsed observables Φ_c become increasingly accurate descriptions of a universal $f_c(z)$ in the limit $t \rightarrow 0$, $L \rightarrow \infty$ with $z = tL^{1/\nu}$ fixed (Eqs. (4)–(13)).

The data are Monte Carlo estimates

$$D = \{(L_i, T_i, m_i, m_i^2, m_i^4, \sigma_{m,i}, \sigma_{m^2,i}, \sigma_{m^4,i})\}_{i=1}^n.$$

For each observation form the collapsed fields and propagate the MC errors,

$$\Phi_{m,i} = m_i L_i^{\beta/\nu}, \quad \sigma_{\Phi_{m,i}} = \sigma_{m,i} L_i^{\beta/\nu}, \quad (15)$$

$$\Phi_{m^2,i} = m_i^2 L_i^{2\beta/\nu}, \quad \sigma_{\Phi_{m^2,i}} = \sigma_{m^2,i} L_i^{2\beta/\nu}, \quad (16)$$

$$\Phi_{m^4,i} = m_i^4 L_i^{4\beta/\nu}, \quad \sigma_{\Phi_{m^4,i}} = \sigma_{m^4,i} L_i^{4\beta/\nu}. \quad (17)$$

Write $\Phi^{(c)} = (\Phi_{c,1}, \dots, \Phi_{c,n})^\top$ for channel $c \in \{m, m^2, m^4\}$, and

$$t_i = \frac{T_i - T_c}{T_c}, \quad z_i = t_i L_i^{1/\nu}. \quad (18)$$

4 Latent model: universal GP + FSS mixture discrepancy

Rather than fitting parametric forms for f_c and g_c , place independent zero-mean GP priors on z ,

$$f^{(c)} \sim \text{GP}(0, k_{\ell_f, \eta_f}), \quad g^{(c)} \sim \text{GP}(0, k_{\ell_g, \sigma_g}), \quad (19)$$

with RBF kernel

$$k_{\ell, \eta}(z, z') = \eta^2 \exp\left(-\frac{(z - z')^2}{2\ell^2}\right). \quad (20)$$

Using the amplitude (14), define the unit-interval mixture gate

$$\pi(t, L) = \frac{a(t, L)}{1 + a(t, L)} \in [0, 1). \quad (21)$$

The collapsed observation model is the *mixture* coupling

$$\Phi_i^{(c)} = (1 - \pi_i) f^{(c)}(z_i) + \pi_i g^{(c)}(z_i) + \varepsilon_i^{(c)}, \quad \varepsilon_i^{(c)} \sim \mathcal{N}(0, \sigma_{\Phi_c, i}^2), \quad (22)$$

where $\pi_i = \pi(t_i, L_i)$. Channels share $(\omega, \kappa, \ell_f, \eta_f, \ell_g, \sigma_g)$ but have independent draws of $(f^{(c)}, g^{(c)})$. There is no extra $L^{k\beta/\nu}$ factor on a in Φ -space: after collapsing, (14) already multiplies the correction channel.

As $t \rightarrow 0$ and $L \rightarrow \infty$, $a \rightarrow 0$ so $\pi \rightarrow 0$ and (22) recovers the plain universal model $\Phi = f + \varepsilon$. When a is large (small L and/or large $|z|$), $\pi \rightarrow 1$ and the point is explained by g instead of f : out-of-window observations do not train the universal scaling function. When $\kappa = 0$, (14) reduces to pure $L^{-\omega}$ gating,

$$\pi = \frac{L^{-\omega}}{1 + L^{-\omega}}, \quad (23)$$

the mixture analogue of the leading Wegner correction in (13).

5 Prior

Collect the inferred scalars into

$$\psi = (T_c, \nu, \beta, \omega, \kappa, \ell_f, \eta_f, \ell_g, \sigma_g) \quad (24)$$

(components that are held fixed are simply omitted from ψ). The prior factorises:

$$\begin{aligned} p(\psi) &= p(T_c) p(\nu) p(\beta) \\ &\quad \times p(\omega) p(\kappa) \\ &\quad \times p(\ell_f) p(\eta_f) p(\ell_g) p(\sigma_g). \end{aligned} \quad (25)$$

Explicitly (matching the defaults used in the code),

$$T_c \sim \text{Uniform}(2.0, 2.5), \quad (26)$$

$$\nu \sim \text{Uniform}(0.5, 1.5), \quad (27)$$

$$\beta \sim \text{Uniform}(0.05, 0.25), \quad (28)$$

$$\omega \sim \text{Uniform}(1, 10), \quad (29)$$

$$\kappa \sim \text{Uniform}(0, 20), \quad (30)$$

$$\ell_f \sim \text{LogNormal}(\log \ell_f^*, \sigma_\ell), \quad (31)$$

$$\eta_f \sim \text{HalfNormal}(\sigma_{\eta_f}), \quad (32)$$

$$\ell_g \sim \text{LogNormal}(\log \ell_g^*, \sigma_\ell) \quad (\text{or tied to } \ell_f), \quad (33)$$

$$\sigma_g \sim \text{HalfNormal}(1), \quad (34)$$

where ℓ_f^*, ℓ_g^* are data-dependent length-scale guesses (fractions of the provisional z -span) and $\sigma_\ell, \sigma_{\eta_f}$ are fixed prior scales. Independently of ψ , the latent fields have GP priors

$$p(\{f^{(c)}, g^{(c)}\}_c \mid \psi) = \prod_{c \in \{m, m^2, m^4\}} \text{GP}(f^{(c)}; \ell_f, \eta_f) \text{GP}(g^{(c)}; \ell_g, \sigma_g). \quad (35)$$

6 Likelihood

Conditional on the latents and ψ , channels are independent and Gaussian:

$$p(D \mid \{f^{(c)}, g^{(c)}\}_c, \psi) = \prod_{c \in \{m, m^2, m^4\}} \prod_{i=1}^n \mathcal{N}(\Phi_i^{(c)} \mid (1 - \pi_i) f^{(c)}(z_i) + \pi_i g^{(c)}(z_i), \sigma_{\Phi_c, i}^2). \quad (36)$$

Integrating out $(f^{(c)}, g^{(c)})$ analytically gives the marginal likelihood used for inference:

$$p(D \mid \psi) = \prod_{c \in \{m, m^2, m^4\}} \mathcal{N}(\Phi^{(c)} \mid 0, K^{(c)}(\psi)), \quad (37)$$

with channel covariances

$$K_{ij}^{(c)}(\psi) = (1 - \pi_i)(1 - \pi_j) k_{\ell_f, \eta_f}(z_i, z_j) + \pi_i \pi_j k_{\ell_g, \sigma_g}(z_i, z_j) + \sigma_{\Phi_c, i}^2 \delta_{ij}, \quad (38)$$

where $\pi_i = \pi(t_i, L_i)$ from (21). Writing this out,

$$\begin{aligned} \log p(D \mid \psi) = \sum_{c \in \{m, m^2, m^4\}} \left[-\frac{1}{2} \Phi^{(c)\top} (K^{(c)})^{-1} \Phi^{(c)} \right. \\ \left. - \frac{1}{2} \log \det K^{(c)} - \frac{n}{2} \log(2\pi) \right]. \end{aligned} \quad (39)$$

(If a channel is disabled, omit its term.)

7 Posterior

Bayes' rule then gives

$$p(\psi \mid D) \propto p(\psi) p(D \mid \psi). \quad (40)$$

Sampling targets ψ with the latent GPs integrated out; predictions for $f^{(c)}$ and $g^{(c)}$ are recovered from the usual GP posterior conditional on each draw of ψ .

References

- [1] K. Harada, *Bayesian inference of the critical exponents from finite-size scaling analysis*, Phys. Rev. E **84**, 056704 (2011).